KAIZEN MATHS

Lesson starter

Taylor and Maclaurin Series

Applications

Name

S1

Use series expansions to evaluate each limit.

Evaluate $\lim_{x \to 0} \frac{1 - \cos x}{x^2}$.

S2

Find the requested coefficient by multiplying the relevant series terms.

Find the coefficient of x^3 in $e^x \sin x$.

S3

Substitute a power series and compare coefficients.

Find the first non-zero terms of the power series solution for $y' = y$, $y(0)=1$.

S4

Use series expansions to evaluate each limit.

Evaluate $\lim_{x \to 0} \frac{e^x - 1 - x}{x^2}$.

Show your working

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Taylor and Maclaurin Series

Applications

Name _____

EA1

Find the requested coefficient by multiplying the relevant series terms.

Find the coefficient of x^4 in $e^x \cos x$.

EA2

Substitute a power series and compare coefficients.

Find the first non-zero terms of the power series solution for $y' = 1 + y$, $y(0)=0$.

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Taylor and Maclaurin Series

Applications

Name _____

EB1

Use series expansions to evaluate each limit.

Evaluate $\lim_{x \to 0} \sin x - x x^3$.

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Taylor and Maclaurin Series

Applications

Name _____

EA1

Find the requested coefficient by multiplying the relevant series terms.

Find the coefficient of x^4 in $e^x \cos x$.

EA2

Substitute a power series and compare coefficients.

Find the first non-zero terms of the power series solution for $y' = 1 + y$, $y(0)=0$.

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Taylor and Maclaurin Series

Applications

Name _____

EB1

Use series expansions to evaluate each limit.

Evaluate $\lim_{x \to 0} \sin x - x x^3$.

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Teacher answers & working

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S1

Use series expansions to evaluate each limit.

Evaluate lim_x→0 1 - cos x x^2 .

Answer:

$\frac{1}{2}$

Working:

Use $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots$.

Substitute into the numerator: 1 - cos x = 1 - (1 - $\frac{x^2}{2}$ + $\frac{x^4}{24}$ - ...).

Simplify the numerator: 1 - cos x = $\frac{x^2}{2}$ - $\frac{x^4}{24}$ + ...

Divide each term by x^2: $\frac{1-\cos x}{x^2}$ = $\frac{1}{2}$ - $\frac{x^2}{24}$ + ...

As x approaches 0, the terms containing x approach 0, so the limit is $\frac{1}{2}$.

S2

Find the requested coefficient by multiplying the relevant series terms.

Find the coefficient of x^3 in $e^x \sin x$.

Answer:

$\frac{1}{3}$

Working:

Write $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$.

Write $\sin x = x - \frac{x^3}{3!} + \dots$.

For x^3, multiply only term pairs whose powers add to 3.

Coefficient = $-\frac{1}{3!} + \frac{1}{2!}$.

Simplify the coefficient: $\frac{1}{3}$.

The product series begins $x + x^2 + \frac{1}{3}x^3 - \frac{1}{30}x^5 + O(x^5)$.

The coefficient of x^3 is $\frac{1}{3}$.

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S3

Substitute a power series and compare coefficients.

Find the first non-zero terms of the power series solution for y' = y, y(0)=1.

Answer:

$y = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots$

Working:

Let $y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \ldots$.

Then $y' = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + \ldots$.

Set y' equal to y and compare matching powers of x.

Constant terms: $a_1 = a_0$.

x terms: $2a_2 = a_1$.

x^2 terms: $3a_3 = a_2$.

x^3 terms: $4a_4 = a_3$.

Since y(0)=1, $a_0 = 1$.

So $a_1 = 1$, $a_2 = \frac{1}{2!}$, $a_3 = \frac{1}{3!}$, $a_4 = \frac{1}{4!}$.

Therefore $y = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots$.

S4

Use series expansions to evaluate each limit.

Evaluate $\lim_{x \to 0} e^x - 1 - x\,x^2$.

Answer:

$\frac{1}{2}$

Working:

Use $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \ldots$.

Substitute into the numerator: $e^x - 1 - x = (1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \ldots) - 1 - x$.

Simplify the numerator: $e^x - 1 - x = \frac{x^2}{2} + \frac{x^3}{6} + \ldots$.

Divide each term by x^2: $\frac{e^x - 1 - x}{x^2} = \frac{1}{2} + \frac{x}{6} + \ldots$.

As x approaches 0, the terms containing x approach 0, so the limit is $\frac{1}{2}$.

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EA1

Find the requested coefficient by multiplying the relevant series terms.

Find the coefficient of x^4 in $e^x \cos x$.

Answer:

$-\frac{1}{6}$

Working:

Write $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$.

Write $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$.

For x^4, multiply only term pairs whose powers add to 4.

Coefficient = $\frac{1}{4!} - \frac{1}{2!2!} + \frac{1}{4!}$.

Simplify the coefficient: $-\frac{1}{6}$.

The product series begins $1 + x - \frac{1}{3}x^3 - \frac{1}{6}x^4 - \frac{1}{30}x^5 + O(x^6)$.

The coefficient of x^4 is $-\frac{1}{6}$.

EA2

Substitute a power series and compare coefficients.

Find the first non-zero terms of the power series solution for $y' = 1 + y$, $y(0)=0$.

Answer:

$y = x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$

Working:

Let $y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots$.

Then $y' = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \dots$.

From $y(0)=0$, $a_0 = 0$.

Substitute into $y' = 1 + y$: $a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \dots = 1 + a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$.

Constant terms: $a_1 = 1 + a_0 = 1$.

x terms: $2a_2 = a_1$, so $a_2 = \frac{1}{2!}$.

x^2 terms: $3a_3 = a_2$, so $a_3 = \frac{1}{3!}$.

x^3 terms: $4a_4 = a_3$, so $a_4 = \frac{1}{4!}$.

Therefore $y = x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$.

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EB1

Use series expansions to evaluate each limit.

Evaluate lim_x→0 sin x - x x^3 .

Answer:

$-\frac{1}{6}$

Working:

Use $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$.

Substitute into the numerator: sin x - x = $(x - \frac{x^3}{6} + \frac{x^5}{120} - \dots) - x$.

Simplify the numerator: sin x - x = $-\frac{x^3}{6} + \frac{x^5}{120} - \dots$.

Divide each term by x^3: $\frac{\sin x - x}{x^3} = -\frac{1}{6} + \frac{x^2}{120} - \dots$.

As x approaches 0, the terms containing x approach 0, so the limit is $-\frac{1}{6}$.